



WASTEWATER, INDUSTRIAL WATER, AQUACULTURE

Screens, filters and spares in polypropylene, without a mould.

Perforated panels, screen elements, nozzles, strainers and the plastic parts the maker no longer supplies: in **sintered polypropylene**, from the drawing or from the old part, from one to a few thousand. The material wastewater treatment has always used, in a geometry a mould cannot make.

≥ 2 mm

APERTURE, HOLES AND SLOTS

≤ 30 cm

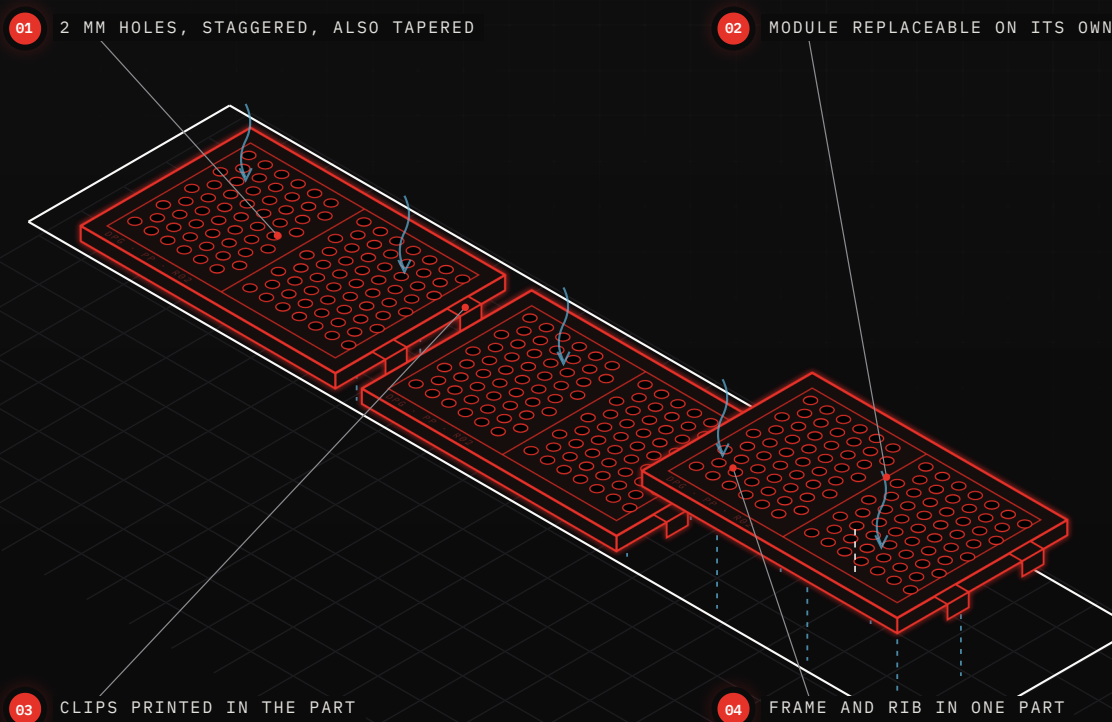
SINGLE PART, MODULES BEYOND

0 moulds

FROM ONE PART TO A SERIES

FIG. 01 - BAND SCREEN PANELS, ISOMETRIC VIEW

TAV. DPG-SET-H20



PERFORATED MODULES CLIPPED TO THE STAINLESS FRAME OF THE SCREEN. IN BLUE: THE WATER

IN RED: THE PARTS MADE BY ADDITIVE MANUFACTURING

01 WHY NOW

2024/3019

EU URBAN WASTEWATER TREATMENT DIRECTIVE

Transposition by July 2027.
Micropollutant (quaternary) treatment mandatory above 150,000 p.e.: 20% of plants by 2033, 60% by 2039, all of them by 2045.

≥ 80%

PAID BY PRODUCERS

Extended producer responsibility: the pharmaceutical and cosmetics industries fund at least 80% of quaternary treatment. New stages, new filters.

Frameworks

SPARES ARE BOUGHT BY TENDER

Thames Water runs a screen services framework covering all makes; in Portugal, SIMARSUL tenders screen spare parts brand by brand.

02 THE SUMS YOU ALREADY KNOW

○ TODAY

The screen panel is ordered from the maker and arrives in weeks, if the model is still in the catalogue. The filter nozzle is from a brand that no longer exists. The strainer is a glued assembly that leaks where it should not, and stainless perforated plate is heavy, costly and needs tooling even for a small variant.



● SINTERED

The old part becomes a file with its revision, and the file becomes a batch: from one part to a few thousand, without a mould. Frame, clips, ribs and thread are born in the same part. The hole can widen towards the outlet so it does not hold, and next time the reorder is identical.

03 PRINTED POLYPROPYLENE IS ALREADY IN THE PLANT

PUBLIC SOURCES, AT THE FOOT OF THE PAGE

WATER UTILITY, UNITED KINGDOM

United Utilities

In its Ofwat-funded Printfrastructure programme it uses printed polymer parts every day: jetting nozzles, instrument troughs and obsolete spares for filter arms.

SCREEN MANUFACTURER

Johnson Screens

Moved the elements of the filtration belt of a wastewater inlet screen from stainless steel to printed polypropylene.

WATER UTILITY, UNITED KINGDOM

Anglian Water

The first part it trialled printing, with the University of Sheffield, was a filter nozzle.

RESEARCH, 2026

PP MBBR carriers

Gyroid biofilm carriers printed in PP, tested on domestic wastewater at pilot scale (Processes, 2026).

● IN PRODUCTION TODAY

For an international water treatment group we produce, as a recurring series, **sintered polypropylene filter panels with a 2 mm aperture**: reordered from a digital warehouse, batch traced. **Not a prototype: a supply.**

⚠ WHY POLYPROPYLENE

It does not absorb water and does not hydrolyse, unlike polyamide. It withstands dilute acids and bases, caustic soda, ferric chloride and cleaning chemicals. It floats (about 0.9 g/cm³) and weighs about an eighth of stainless steel. It is what tanks, pipes and nozzles have always been made of.

SOURCES Directive (EU) 2024/3019, eur-lex.europa.eu · Hydro International, Thames Water screen services framework, hydro-int.com · SIMARSUL, TED notice 056207-2024 · United Utilities, "Printfrastructure", unitedutilities.com · Johnson Screens and HP, johnsonscreens.com; 3dprint.com, 2022 · Anglian Water and Univ. of Sheffield, 3dprint.com, 2015 · Processes 14(6):1001, 2026, doi 10.3390/pr14061001 · Stratasy, SAF PP: material datasheet and watertightness white paper · PP chemical resistance: Braskem, INEOS, Charlotte Pipe

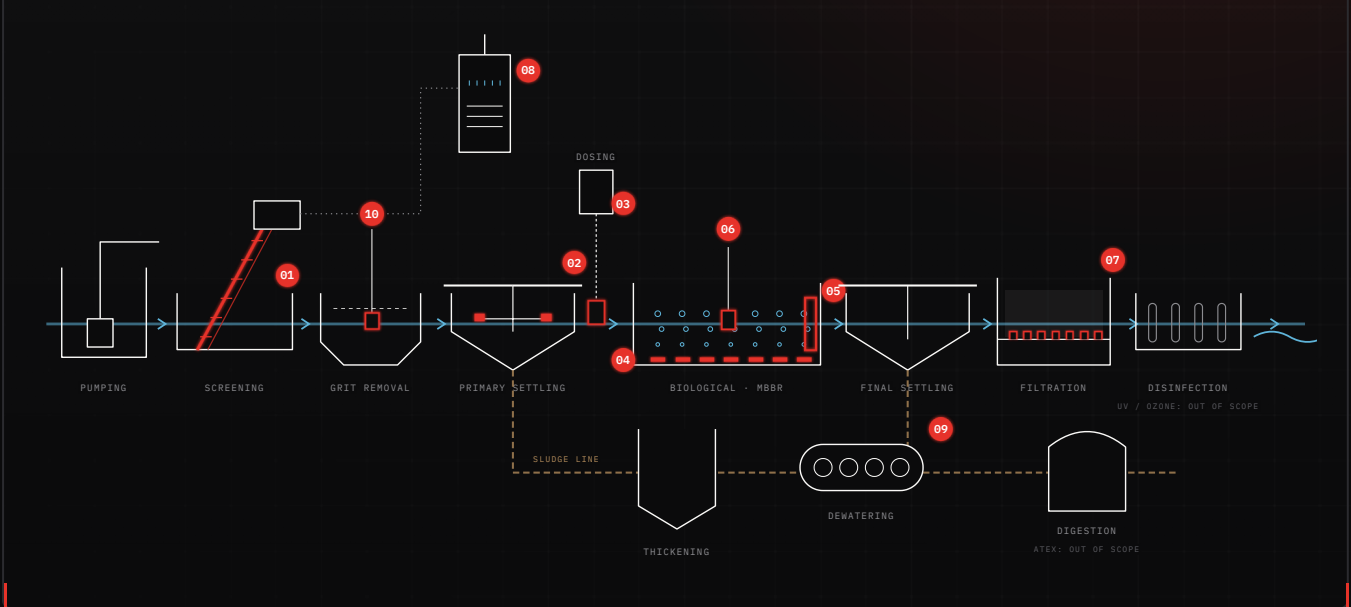


PLANT MAP

From the screens to the sludge: ten places where a printed PP part makes sense.

FIG. 02 - LAYOUT OF AN ACTIVATED SLUDGE PLANT

TAV. DPG-SET-H20



IN BLUE THE WATER, DASHED THE AIR AND THE SLUDGE LINE. FIT: HIGH = PART
ALREADY IN PLASTIC, PLANT-SPECIFIC, HARD TO SOURCE

IN RED: WHERE PRINTED PP COMES IN

- | | |
|---|--|
| 01 Screen panels and elements HIGH
Band screens, rake-comb screens, drum screens: 2 mm perforated panels, elements, guides, wear shoes, brush holders. | 02 Scraper small parts HIGH
Spacers, filler blocks, scum skimmer tips, weir adjustment brackets. |
| 03 Chemical dosing MEDIUM
Small static mixers, foot-valve strainers, injection tips for ferric chloride and hypochlorite, low pressure. | 04 Aeration MEDIUM
Saddles, rings, adapters and fittings for diffuser grids no longer made. The diffuser itself stays standard. |
| 05 MBBR retention sieves HIGH
Modules with 4-5 mm slots and an anti-wedging profile, for small, industrial and pilot plants. | 06 Probe holders HIGH
Cages, pendulum probe holders, shrouds with the air-cleaning nozzle built into the part. |
| 07 Nozzles and strainers HIGH
Second source for off-catalogue models: biofilters, underdrains, ion exchange. 2 mm slots. | 08 Odour control MEDIUM
Liquid distributors and headers with the nozzles in the same part, supports, mist eliminators. |
| 09 Sludge line MEDIUM
Wash-nozzle holders, blade holders, chicanes for belt presses and thickeners. | 10 Instrumentation HIGH
Stilling wells, sampler strainers, brackets across the whole plant. |

OUT OF SCOPE, AND WE SAY SO FIRST

contact with ozone and chlorine dioxide

pump impellers and centrifuges

large GRP baffles and weirs

standard MBBR carriers and packing

fine sand-filter slots below 2 mm

Atex zones of the biogas line

01 FOR UTILITIES, MAINTENANCE TEAMS AND SCREEN SERVICE COMPANIES

The plastic spare you can no longer find.

Panels, elements and guides of band and rake-comb screens, drum screens, covers, wear shoes, brush holders and nozzle holders: we remake them in PP from the old part, also for machines no longer built, and also a single piece. For multi-brand service companies we are the supplier of the parts the maker no longer supplies, or supplies in weeks.



The old part

or the drawing, or the STEP. Even broken, even from a brand that is gone.



3D survey

we measure the part and rebuild dimensions, threads and fits.



Revised file

the spare becomes your own part number, with its revision.



PP batch

from one part to a few thousand, traced, without a mould.



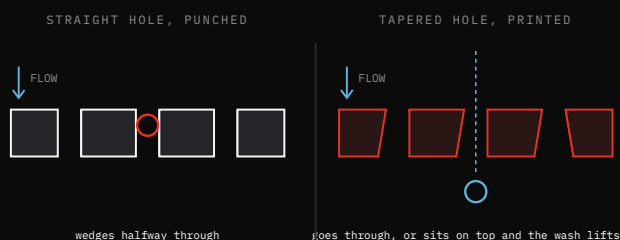
Identical reorder

in one year or in ten, from the file, with no shelf.

02 FOR MAKERS OF SCREENS, DRUM SCREENS AND FILTERS

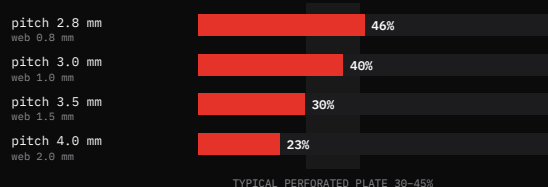
The perforated panel in plastic, instead of the plate.

FIG. 03 - PANEL CROSS-SECTION



In plate, the hole is straight. Printed, it can widen towards the outlet like a wedge-wire profile: what goes in, goes through.

FIG. 04 - OPEN AREA, 2 MM HOLES, STAGGERED



TYPICAL PERFORATED PLATE 39-45%

Geometric calculation. The minimum web is set with you, on the stiffness the panel needs.

	PERFORATED STAINLESS PLATE	INJECTION-MOULDED PANEL	SINTERED PP
TOOLING	punching or laser	one mould per format	none: the file
SENSIBLE BATCH	from one part	thousands	from one part to a few thousand
HOLE SHAPE	straight	whatever can be demoulded	straight, tapered, varied by zone
FRAME AND CLIPS	welded or bolted	in the mould	in the part, even for a single spare
WEIGHT	7.9 g/cm³	light	about 0.9 g/cm³
CHLORIDES, SEAWATER	stainless suffers	no problem	no problem
SAND AND GRIT	good	excellent in polyurethane	fair: downstream of grit removal
MINIMUM APERTURE	below one millimetre	fine	2 mm

- **Frame, ribs, clips and gasket seat** in the same part: the module clips in and is replaced on its own.
- **Different apertures by zone** on the same panel, and the part number engraved on the frame.
- **An eighth of the weight of stainless:** panels are changed by hand, without a hoist.
- **Modules up to 30 cm** a side. Beyond that, the panel is split into modules that clip together, like the polyurethane panels of vibrating screens.
- **Where it does not fit:** sand and grit upstream of grit removal wear PP faster than polyurethane and UHMW; below 2 mm, wedge wire and cloth remain the answer.

03

FOR THOSE WHO MAKE, SELL OR FIT NOZZLES AND STRAINERS

Nozzles and strainers: the second source.

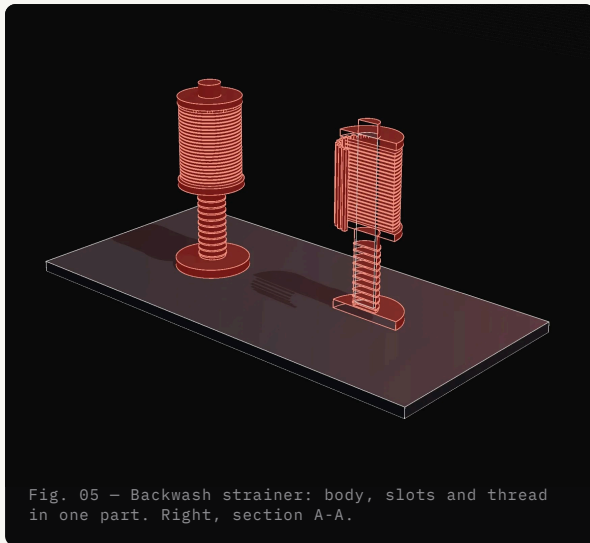


Fig. 05 – Backwash strainer: body, slots and thread in one part. Right, section A-A.

The off-catalogue model, the brand that no longer exists, the variant for a single plant: we make it in PP, from the part or the drawing, with the same thread.

- **Body, slots and thread in one part:** no gluing, no joint to leak.
- **Wedge-shaped slot** widening inwards: it cleans itself at backwash.
- **The air-scour hole** in the stem, in the same part, if your nozzle has one.
- **Slots from 2 mm:** biofilters, gravel underdrains, ion exchange, aquaculture. For fine sand we print the body and the slotted cap stays off the shelf.

04 IDEAS BEYOND THE SCREEN

WHERE THE SAME PP ALREADY WORKS

EVERY PLANT

Self-cleaning probe holders

Cages and shrouds for oxygen, pH and turbidity probes, with the air-cleaning channel inside the part.

limit · low loads, plant-specific parts

COMPACT PLANTS, INDUSTRY, PILOTS

MBBR retention sieves

Flat modules instead of the stainless wedge-wire cylinder, 4–5 mm slots, a profile that does not wedge the carrier.

limit · large plants: in modules

ODOUR CONTROL

Scrubber distributors

Liquid distributor and nozzles in one part, no welds, in PP's own scrubbing chemistry: caustic, hypochlorite, sulphuric acid.

limit · large eliminators: in modules

TERTIARY, AQUACULTURE

Disc and drum filter spares

Wash nozzles and spray bars, suction shoes, panel frames: the cloth stays cloth, the support is printed.

limit · cloth below 2 mm cannot be printed

RAS, HATCHERIES

Recirculating aquaculture

Tank outlet screens sized to the fish, degasser distribution plates, nozzle holders. In seawater stainless suffers, PP does not.

limit · outdoor UV to be assessed

PRE-TREATMENT

Plating and industrial water

Spargers, mixers and internals of neutralisation tanks, where PP is already the house material.

limit · no aromatic or chlorinated solvents

PROCESS WATER

Cooling towers

Spray nozzles and adapters for towers whose spares can no longer be found, with the orifice you need.

limit · above about 60 °C to be assessed

DAIRIES, WINERIES, FISH PROCESSING

Food industry

Spares for micro-screens and static sieves, gully baskets, grease-trap paddles: small parts of machines the maker no longer supports.

limit · not for food contact

GULLIES, OVERFLOWS

Stormwater

Baskets and screens for non-standard gullies, outfall screens, in modules that clip together.

limit · large baskets: in modules

UNIVERSITIES, ENGINEERING FIRMS

Pilot plants and research

Gyroid biofilm carriers, small drum screens, diffuser holders: the prototype that becomes a series.

limit · cost per litre at full scale

**05 THE MATERIAL**

SAF POLYPROPYLENE · DPG-MAT-PP

Tensile strength	30 MPa
Tensile modulus	1600 MPa
Elongation at break	20%
Heat deflection, 0.45 MPa	100 °C
Density	about 0.9 g/cm ³
Watertight	from 1 mm wall

Datasheet values from the powder maker, on test specimens. The part is assessed on the part.

● Dilute non-oxidising acids	sulphuric, hydrochloric
● Bases	caustic soda
● Coagulants	ferric chloride
● Hydrogen peroxide	at room temperature
● Sodium hypochlorite	cold yes; hot and continuous exposure to be assessed
● Polyelectrolytes, PAC, H ₂ S, methanol, peracetic acid	to be checked in your fluid
● Ozone, chlorine dioxide	degrade PP: PVDF is needed
● Concentrated oxidising acids, aromatic and chlorinated solvents	attack or swell PP

● suitable ● with conditions ● no · base-polymer data: on request we test the part in your fluid.

06 MATERIALS FOR THE SECTOR

MATERIAL	PROCESS	WHY HERE	REFERENCE
Polypropylene	SAF	the water material: does not absorb water, does not hydrolyse, withstands process chemistry; UL94 HB, not for fire requirements	DPG-MAT-PP
PA2200 (PA12)	SLS	where more stiffness or temperature is needed, up to +80 °C under load, out of continuous immersion: polyamide absorbs water	DPG-MAT-PA2200

07 THE FLIP SIDE

WHEN IT DOES NOT PAY, WE TELL YOU FIRST

Below 2 mm aperture

the fine slot of sand filtration stays with wedge wire and cloth.

Drinking water

our PP holds no approval for water intended for human consumption.

Sand and grit

upstream of grit removal, polyurethane and UHMW last longer.

Cold and sun

below 0 °C PP becomes brittle on impact; outdoors, UV resistance is assessed.

Beyond 30 cm

the part is split into modules; pipes, sheets and large baffles are not our trade.

Ozone and chlorine dioxide

PP does not withstand them: that stage needs PVDF.

Pressure over time

we do not state a pressure rating: it is tested on your case.

FOR A QUOTE WE NEED 1. The old part, even broken, or the drawing or STEP 2. Two photos of the part in place
3. The fluid, the chemistry and the operating temperature 4. How many parts: now, and per year

Send us the part you can no longer find.
We also tell you when it does not pay.

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